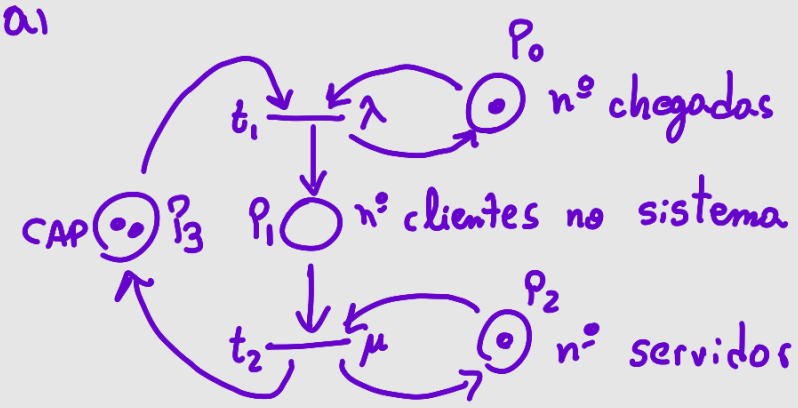
①

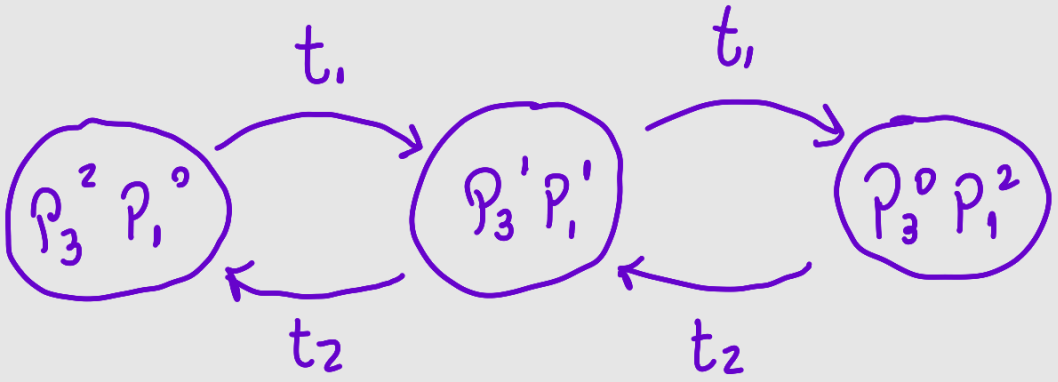
$\lambda = 1\,min^{-1}$

$\frac{1}{\mu} = 1\,min \rightarrow \mu = 1\,min^{-1}$

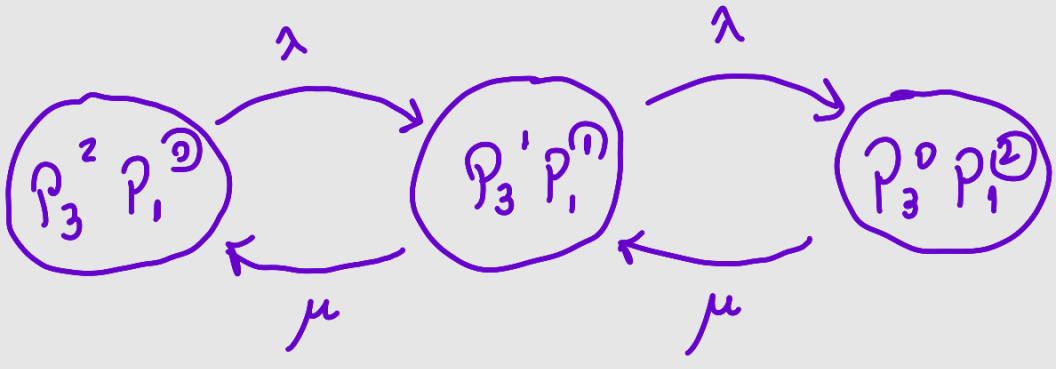
a)

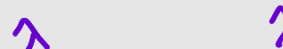


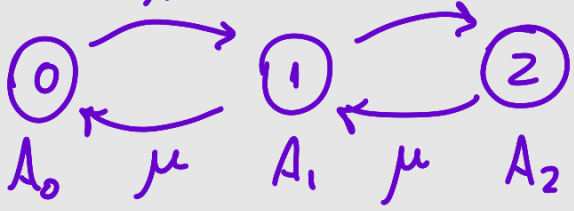
b)



c)







d)

Est 0: $\lambda \cdot A_0 = \mu A_1$ $\quad A_0 = A_1 = A_2 = 1/3$

Est. 2: $\lambda \cdot A_1 = \mu \cdot A_2$

Norm.: $A_0 + A_1 + A_2 = 1$

Lei de Little: $\bar{n} = \lambda \cdot \bar{\sigma}$ → Vale p/ todo processo nasc. morte

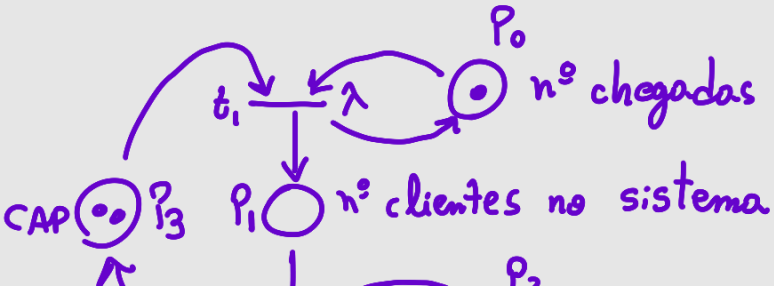
$\bar{n} = 0 \cdot A_0 + 1 \cdot A_1 + 2 \cdot A_2$

$\bar{n} = 1$

$\bar{\sigma} = \frac{\bar{n}}{\lambda} \rightarrow \bar{\sigma} = 1\,min$

② M/M/2/2

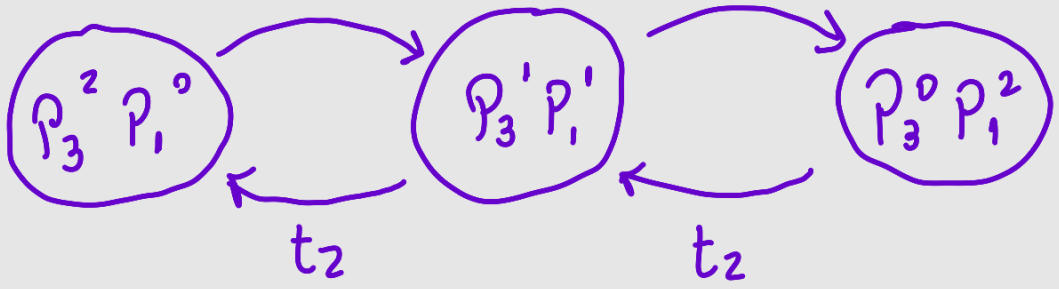
a)



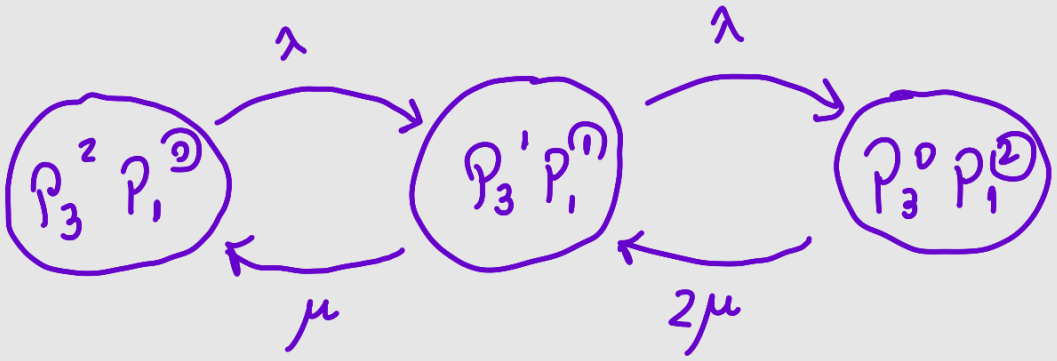


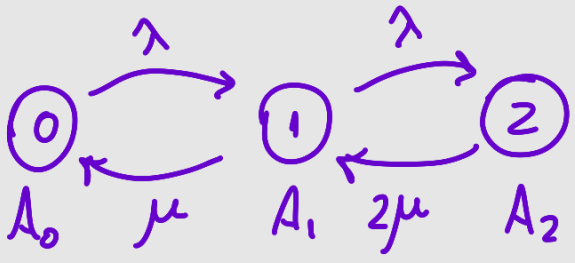
b)

$$\begin{bmatrix} 1 \\ 0 \\ 2 \\ 2 \end{bmatrix} \begin{matrix} P_0 \\ P_1 \\ P_2 \\ P_3 \end{matrix}$$



c)





d) $P_0 = 0,4$
$P_1 = 0,4$
$P_2 = 0,2$
$\}$ (0,8) → 80%.

3

